

Program : Diploma in Printing Technology	
Course Code : 4102	Course Title: Print Finishing & Conversion Techniques
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To provide the basic knowledge on book binding and print finishing operations doing in post press.
- To guide the student to know about the importance and need of doing print finishing operations in press work.
- To familiarize with all the book binding aspects.
- To identify and understand post press practices and related machineries.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Paper sizes, Paper Grades		Materials for Printing & Packaging.	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the role and necessities of book binding and print finishing operations.	15	Understanding
CO2	Explain the key features and operations of book binding.	14	Understanding
CO3	Explain about forwarding and finishing operations of post-press.	15	Understanding
CO4	Describe the key features and advancements of post press machineries.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the role and necessities of book binding and print finishing operations.		
M1.01	Introduction to Book binding and Print finishing.	3	Understanding
M1.02	Different types of Book Binding.	4	Understanding
M1.03	Binding & Finishing Tools and Equipments.	4	Understanding
M1.04	Binding and Finishing Materials.	4	Understanding
Contents: Definition - Book binding and Print finishing. Classification of binding-Stationery Binding and Letter press Binding and its subdivisions. Loose Leaf Binding - Introduction, Types - Spiral Binding, Ring, Thong, Post and Comb Binding. Perfect Binding - Introduction, Features and advantages. Case Binding - Introduction, Features and advantages. Account Book Binding - Definition, Features and advantages. Binding & Finishing Tools and Equipments. Binding Materials - Covering Materials, Reinforcing Materials, Securing Materials, Adhesives. Print Finishing Materials-Eyelets, Foil and Coloring material.			
CO2	Explain the key features and operations of book binding.		
M2.01	Stitching and sewing process done in book binding.	4	Understanding
M2.02	Folding schemes/styles	3	Understanding
M2.03	Paper Sizes		Understanding
M2.04	Styles of binding	3	Understanding
M2.05	End Paper	4	Understanding
	Series Test I	1	

Contents:

Stitching - Definition, Types-Side Stitching and Saddle Stitching. Sewing - Definition, Types - Overcast Sewing and All along Sewing- French Sewing, Tape Sewing or Sewing on Tapes, Raised Cord Sewing or Flexible Sewing, Recessed Card Sewing or Sawn-in Sewing, Two on sewing. Paper sizes and Folding schemes-Folio,Quartavo,Oactavo,16mo and 32mo,64mo.Folding styles: Right Angle Folding or French fold & Parallel Folding-Zig zag fold, Letter folds, over and over fold, Styles of binding - Quarter bound book, Three Quarter bound book, Half bound book, Full bound book. End Paper-Definition, purpose of use, Types-Self end paper, Single End Paper, Double end paper, Made End Paper, Reinforced End Paper, Cloth Joint End Paper, Zig-Zag End paper.

CO3	Explain about various forwarding and finishing operations of post press.		
M3.01	Pre-Forwarding and Forwarding operations done in post press.	3	Understanding
M3.02	Basics of various Pre-forwarding operations.	3	Understanding
M3.03	Basics of various Forwarding operations.	3	Understanding
M3.04	Major Print finishing operations done in post press.	3	Understanding
M3.05	Basics of various print finishing operations.	3	Understanding

Contents:

Pre-Forwarding and Forwarding Operations - Overview and sequencing of operations. Definition, types, working principle and purpose of use of - Cutting & Trimming, Folding, Gathering, Collating, Sewing, Rounding and Backing, Head banding, Casing in, Creasing, Slitting, Punching, Drilling. Print Finishing Operations - Definition, Purpose. Definition and basic working principle of - Embossing, Debossing, Gold Blocking /Foil Stamping, Ruling, Lettering, Eyeleting, Laminating, Coating, Varnishing, Index cutting, Inserting, Tipping-in, Numbering, Perforating, Die cutting and Edge Decoration operations such as Edge Coloring, Painting, Sprinkling, Gaufering, Marbling , Edge Gilding etc.

CO4	Describe the key features and advancements of post press machineries.		
M4.01	Post press machines and their basic functions.	4	Understanding
M4.02	Components & advanced features and general working principle of major post press machines.	5	Understanding
M4.03	Modernization & Advancements in post-press.	3	Understanding
M4.04	Quality control in post-press operations.	2	Understanding
	Series Test II	1	

Contents:

Component description, advanced features and general working principle of-Programmable Cutting Machine, Folding Machine, Wire Stitching Machine, Rounding and Backing Machine, Gathering Machine, Laminating machine, Coating machine, Bundling machine, Perfect binding machine, Case making machine. Modernization & Advancements in post-press. Quality control in post-press operations.

Text / Reference:

T/R	Book Title/Author
T1	D. Mendiratta, Binding & Finishing , Printek Publication, New Delhi 1999.
R1	Spiers Hugh, "Introduction to Printing and Finishing" , PIRA UK, 1988.
R2	G. Martin, (1980), Finishing process in Printing Focal Press , London.
R3	Alex J. Vaughan- Modern Book Binding .
R4	Ralp Lyman, (1993), Binding and Finishing, GATF Press .
R5	Folding in Practice – by Alfred Furler.
R6	P. Kipphan, Handbook of Print Media , Springer, 2002.

Online resources:

Sl. No	Website Link
1.	http://printwiki.org/Binding_and_Finishing
2.	https://en.wikipedia.org/wiki/Bookbinding
3.	https://post-press.com/glossary-of-printing-terms/
4.	https://printninja.com/printing-resource-center/printing-academy/post-press